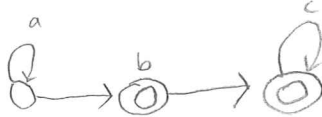


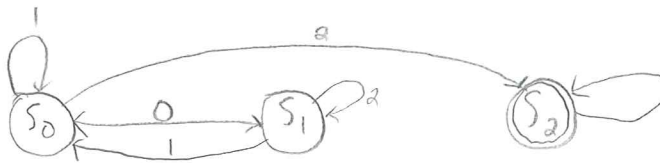
Name Brandon Buchanan Finite State Machines

1) Draw the digraph for FSM for the regular expression a^*bc^*

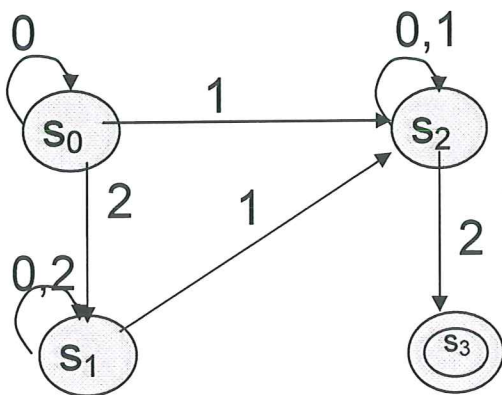


	0	1	2
S ₀	S ₁	S ₀	S ₂
S ₁	S ₀	S ₀	S ₁
S ₂	S ₂	S ₂	S ₂

2) Draw the digraph of the machine from the following state transition table.



3) Construct the state transition table of the FSM for the following digraph and give the regular expression describing the FSM.



	0	1	2
S ₀	S ₀	S ₂	S ₁
S ₁	S ₁	S ₂	S ₁
S ₂	S ₂	S ₂	S ₃
S ₃	-	-	-

$$0^*1(0,1)^*2 \vee 0^*2(0,2)^*1(0,1)^*2$$